**Program 1: Getting Started - Tableau Workspace, Tableau terminologies, Basic functionalities.**

1. Go to Start Page: Toggle between the active sheet and the Desktop Start Page.
2. Data Pane: Includes dimensions and measures, populated from your selected data source. May also include calculated fields, parameters, or sets.
3. Analytics Pane: Includes options you can use to apply reference lines, forecasts, trend lines, to add totals to crosstabs, and to build boxplots.
4. Workbook Name: The file name of our workbook.
5. View Cards: Used for modifying the worksheet.
6. Toolbar Icons: Icons are available for quick access to popular features.
7. Worksheet/View: Workspace for building your visualizations.
8. Go to Data Source: Returns you to the data source specification page.
9. Worksheet Tabs: Click to view a specific worksheet, dashboard, or story
10. New Worksheet, Dashboard, and Story Tabs: Click to create a new Worksheet, Dashboard, or Story.
11. Status Bar: Displays data about the fields and marks included in the view.

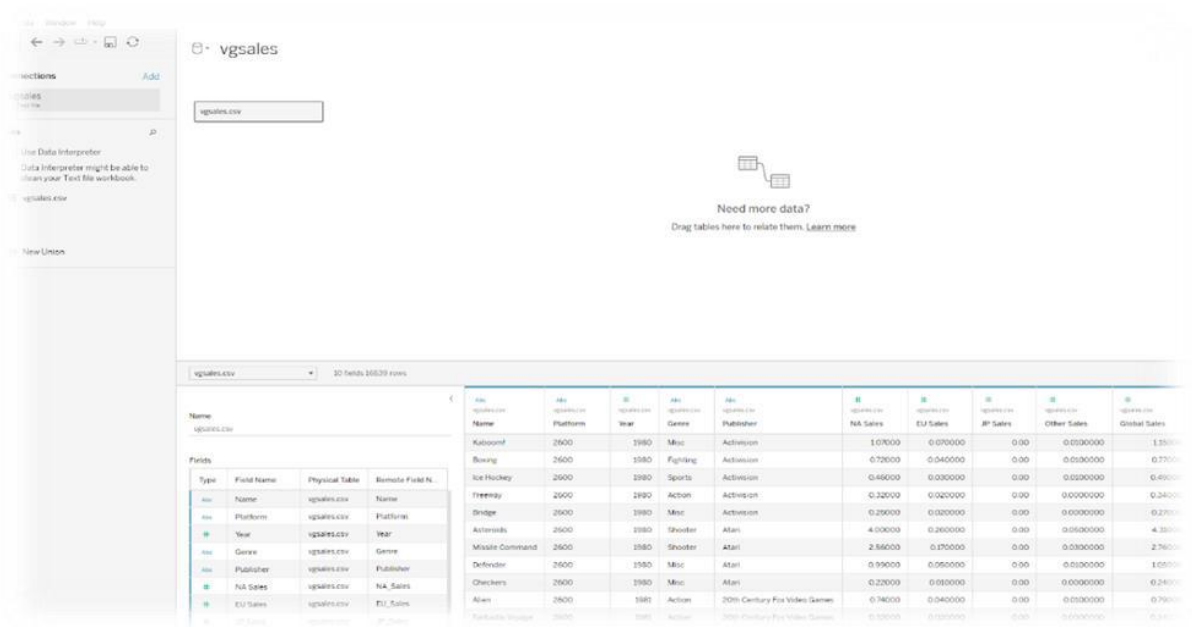
**Steps:**

1. **Tableau Workspace Setup:**

- **Connect to Data:**
- Open Tableau, and on the "Start Page," select Connect -> To a File -> Text File.
- Browse to the location of vgsales.csv and open it

**Data Preview:**

- After loading, Tableau will show a preview of the data. You can rename columns if necessary.
- Click on the "Sheet 1" tab at the bottom to go to your first worksheet.




2. **Tableau Terminologies:**

- **Dimensions:** These are qualitative fields. In vgsales.csv, examples include Platform, Genre,and Publisher.
- **Measures:** These are quantitative fields used for calculations. Examples are Global_Sales, NA_Sales, and Year.
- **Rows and Columns Shelf**: Drag dimensions and measures to the Rows or Columns shelves to build the structure of your visualization.
- **Marks:** Controls the appearance of the data. You can set marks to be circles, bars, or other shapes and control size, color, and label.
- **Filters:** Used to limit the data displayed in the view.
- **Pages Shelf:** Used for creating animations or segmenting your view by categories.

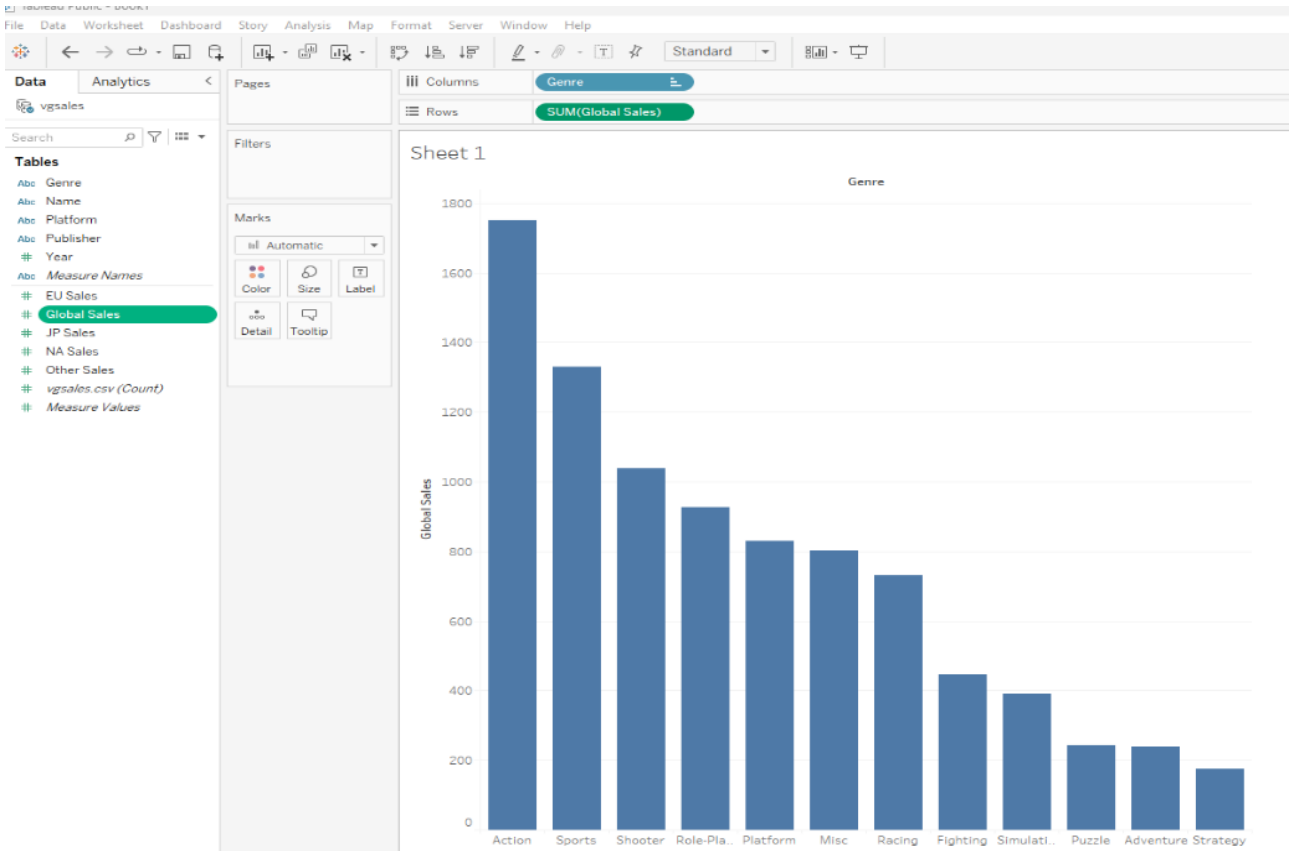
## 3. Basic Functionalities:

### a. Basic Visualization (Bar Chart of Global Sales by Genre):

- In your worksheet, drag Genre to the Columns shelf.
- Drag Global_Sales to the Rows shelf.
- You should see a bar chart. If the data isn't aggregating correctly, check if the aggregation is set to SUM by right-clicking Global_Sales -> Measure -> Sum.

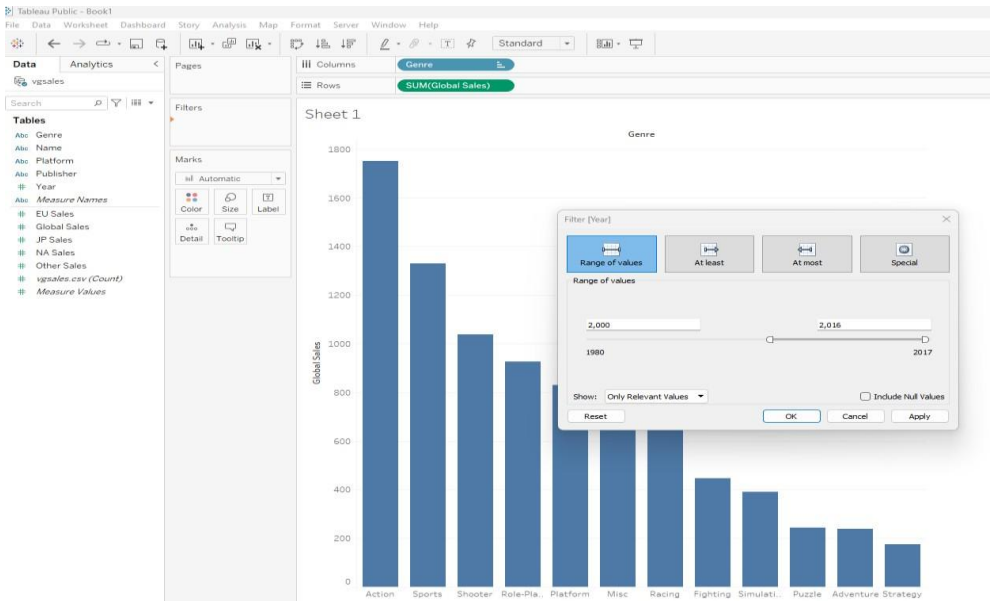
### b. Sorting:

- Click on the Global_Sales axis and sort descending to show the genres with the most sales first.



### c. Filtering:

- Drag Year to the Filters shelf.
- Choose the range of years you want to display (e.g., 2000-2015).
- Add Year to the Pages shelf to create a dynamic view of how sales changed over time.

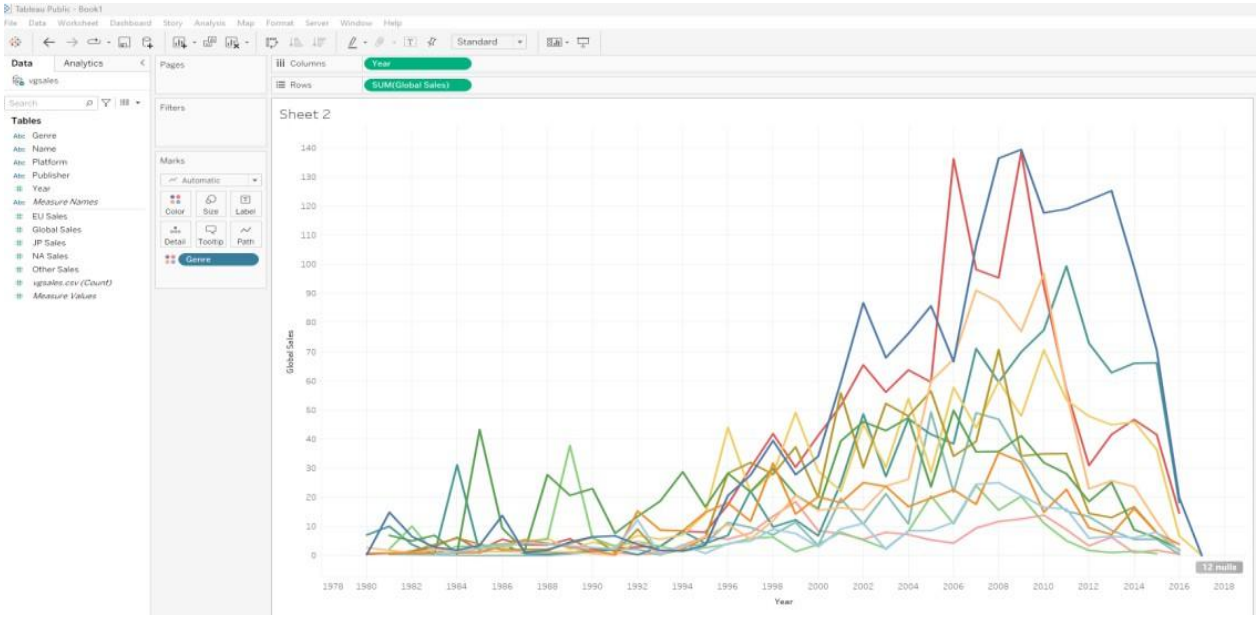


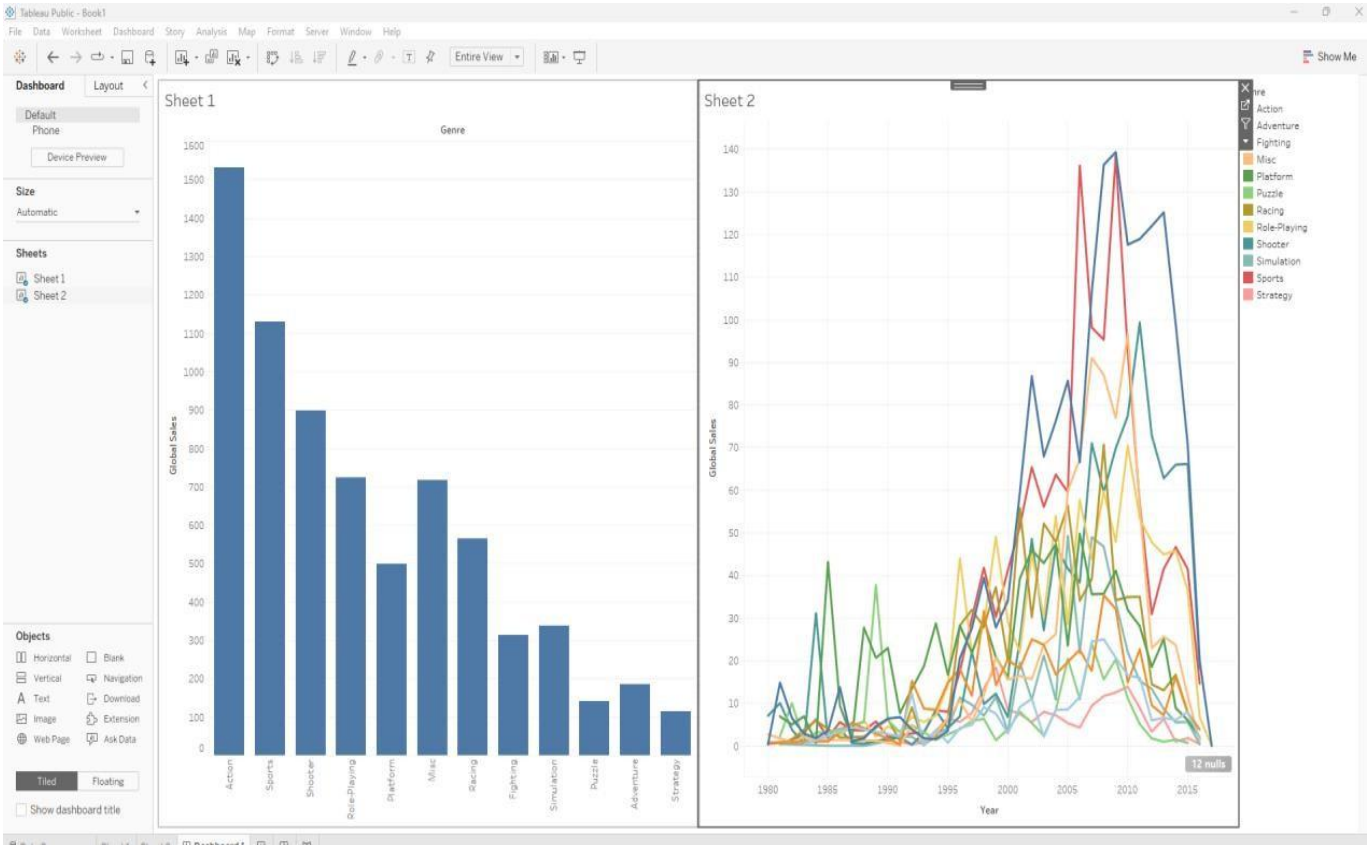

### 4. Additional Functionalities:

**Dashboards:** Combine different sheets to create a comprehensive dashboard. Go to the Dashboard tab, drag your created sheets to the layout, and arrange them accordingly.

### a. Add one more worksheet - Global Sales Trend by Year

- Drag Year to the Columns.
- Drag Global_Sales to the Rows.
- Create a line chart to show how global sales have trended over time.
- Add Genere to the Marks
- Apply color to Genere




b. Go to the Dashboard tab in Tableau.
c. Add multiple visualizations to a single dashboard. (To increase dashboard size - select size-Automatic)
d. Arrange charts (e.g., a line chart for yearly sales, a bar chart for top genres, etc.).